CLASS 10 OUR ENVIRONMENT NOTES BY

Study Root

CH 13 Science - Notes

Differences btw	Environment and Ecosystem
Environment	**Ecosystem**
• All those things and set of conditions which influence life of an organism is called Environment.	Community where biotic and abiotic components interact with each other is called an Ecosystem.
• Environment changes as we move from place to place	• Ecosytem remains constant

Components of Ecosystem

Ecosystem consists of two components biotic and abiotic . Biotic components include all those living organisms present in the ecosystem . It is divided into three categories -

- **Producers** - Those organisms who make their own food and are also called autotrophs. e.g. plants , blue-green algae
- **Consumers** - Includes those organisms who depend on producers for their food either indirectly or directly, e.g. lions

- **Decomposers** - These are microorganisms that feed on dead and decaying matter breaking complex substances into simple inorganic substances, making it once again available for plants.

Abiotic components are non-living components of an ecosystem on which an organism depends. It includes factors like -

- Light
- Temperature
- Atmospheric gases
- Wind
- Water, etc.

There are two types of ecosystem. The first one is Natural and the other is Artificial ecosystem.

- **Natural Ecosystem** - Naturally existing ecosystem without any human support.

  E.g. Aquatic - ponds , lakes , etc.

  Terrestrial - desert , grassland , forest, etc.

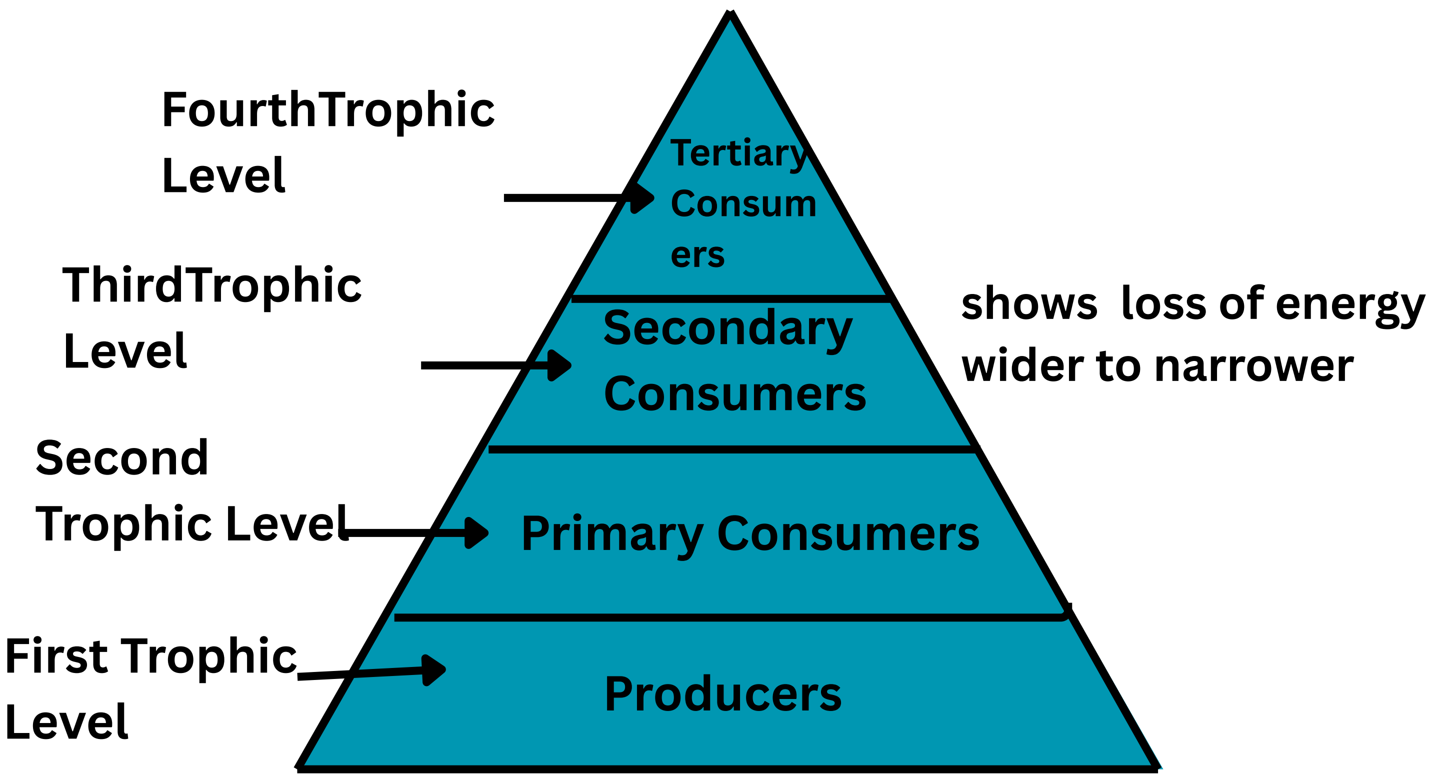
- **Artificial Ecosystem / Man-made Ecosystem** - Those ecosytems which are created and maintained by humans.

  E.g. Aquariums , Crop fields , gardens , etc.

Food Chain - Food chain is linear flow of energy and nutrients from one organism to another .

Food Web - Food web is network of various food chains being interlinked.

Trophic level - Trophic level is position of an organism in a food chain . Each organism in a food chain stands on a particular trophic level.



10 % Energy Law- When one form of energy is converted to another some energy is lost to the environment in the form which can not be used again Thus energy available at each trophic level gets diminished progressively.

An average of 10% of food eaten by an organism is turned into its own body and made available for next levels of consumers. Therefore , 10 % can be taken as average value for the amount of organic matter that is present at each step and reaches next level of consumers. This is called

Law.

(**NOTE** : Green plants in a terrestrial ecosystem capture about 1% of energy that falls on their leaves and convert into food energy . Not 10 %)

Food chains generally consist of only three to four trophic levels because loss of energy is so great that a little energy remains available for next level of consumers and after four trophic level very little usable energy remains.

Biomagnification or Biological Magnification is the phenomenon of an increase in concentration of toxicant at each successive trophic level.

E.g., Use of chemicals to protect crops → They are often washed down into the soil

↓

from soil absorbed by plants ← accumulates in our body as we occupy top level in any food chain

Ozone

Ozone is a molecule formed by three atoms of oxygen and is a deadly poison , but in stratosphere it shields the surface of earth from UV (ultraviolet) radiation from the sun.

Formation - UV radiation splits apart some molecular oxygen (O2) into free oxygen (O) atoms which combine with molecular oxygen to form ozone.

Amount of ozone dropped sharply in 1980s because of synthetic chemicals like CFCs (chlorofluorocarbons) which were used in refrigerants and in fire extinguishers.

In 1987 UNEP (United Nations Envrionment Programme) put an agreement to freeze CFCs production at 1986 levels. It is now mandatory for all manufacturing companies to make CFC-free refrigerators.

Waste - Left or discarded substances . Can be either gaseous , liquid or in solid form.

Differences Btw BIodegradable and	Non-Biodegradable Waste
Substances that can be broken down into harmless simpler forms naturally by the action of microorganims are called Biodegradable waste.	Substances that can not be broken down into harmless simpler forms by the action of microorganisms are called Non-Biodegradable waste.
It is generally environment-friendly.	It is harmful for the environment and causes environmental pollution.
Can be converted into manure or fertiliser	Can be either recycled or reused

Methods of Waste Disposal

- Recycling
- Composting
- Incineration
- Seawage Treatment
- Landfill
- Biogas Production

Garbage - Household waste

Imp

Effect of Biodegradable waste on the environment

- Their decomposition leads to a foul smell .
- Dumping of industrial waste reduces soil fertility thus affecting crop yield.
- Dumping of waste into water bodies leads to water pollution, causing water-borne diseases.

Imp

Effect on Non-Biodegredable waste on the environment

- They pollute water and harm aquatic life.
- Toxic metal wastes like lead , mercury , etc., accumulate in environment causing life-threatening diseases.

THANK YOU!